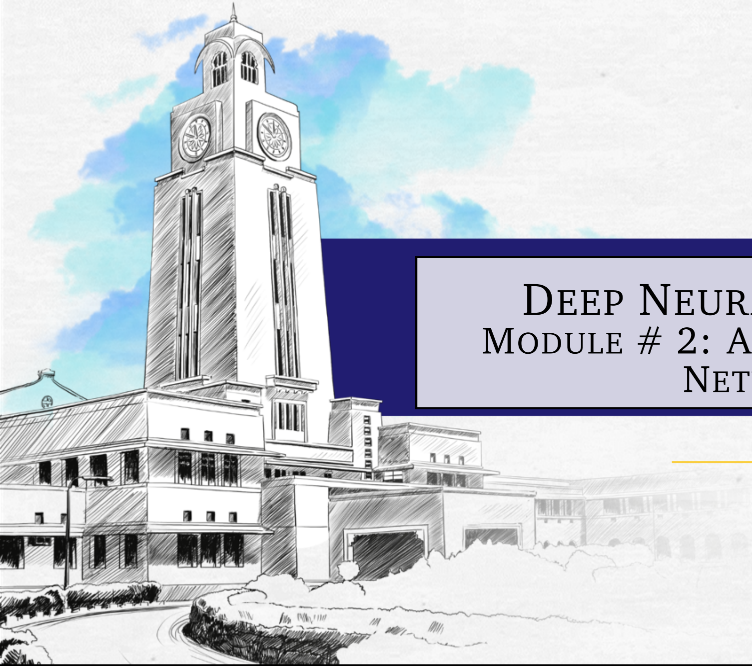

Deep Neural Network

Module # 2: Artificial Neural Network

DNN Team

BITS Pilani WILP

With grateful acknowledgement to the authors who have generously made their course materials freely available online.
Deck prepared by Seetha Parameswaran.

Questions that will be answered

1. What is an artificial neuron?
2. How is it related to biological neuron?
3. How are artificial neurons interconnected?
4. How do artificial neuron learn?

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How do Humans Learn?

It begins with brain.

- Humans learn, solve problems, recognize patterns, create, think deeply about something, meditate and many many more....
- Humans learn through association. [Refer to Associationism for more details.]

OBSERVATION: THE BRAIN

- The brain is a mass of interconnected neurons.
- Number of neurons is approximately $\boxed{10^{10}}$.
- Connections per neuron is approximately $\boxed{10^{(4\ to\ 5)}}$.
- Neuron switching time is approximately $\boxed{0.001 \text{ second}}$.
- Scene recognition time is 1 second.
- 100 inference steps doesn't seem like enough.
- Lot of parallel computation.

Brain: Interconnected Neurons

- Many neurons connect in to each neuron.
- Each neuron connects out to many neurons.

Biological Neuron Structure

Input Signal

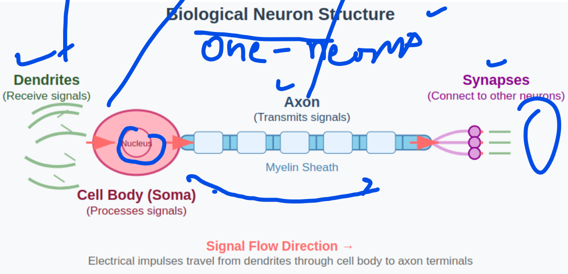
Process

O/P Signal

Key Components:

- Dendrites: Receive signals from other neurons
- Cell Body (Soma): Processes incoming signals
- Axon: Transmits output signal
- Synapses: Connection points between neurons

One - Neuron



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ANN

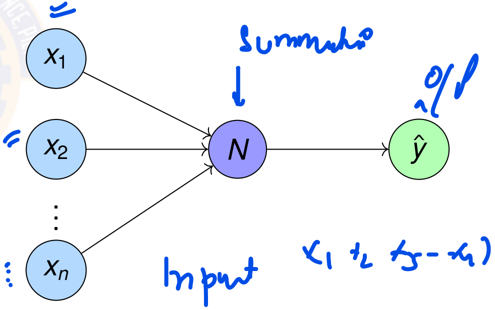
Biological	Artificial
1. Dendrites	Input
2 Synapses	Weight
3. Cell body	Summation
4 Axon	Output

What is an Artificial Neuron?

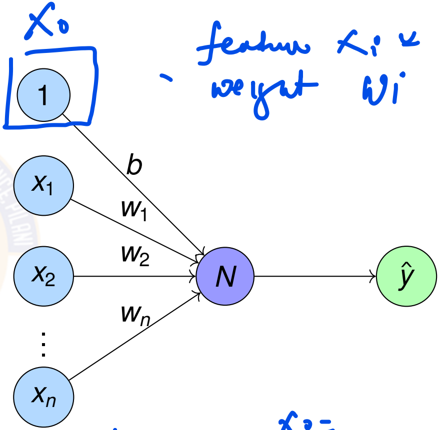
Definition

Artificial Neuron is a processing element inspired by how the brain works.

- Similar to biological neuron, each artificial neuron will be do some computation. Each neuron is interconnected to other neurons.
- Similar to brain, the interconnections between neurons store the knowledge it learns. The knowledge is stored as parameters.



- Processing neuron N performs weighted sum.
- For each feature x_i, weight w_i shows the importance to the feature .
- N generates one numerical output $\hat{y}$



feature x_i ↔ weight w_i

$x_1w_1 + x_2w_2 + x_3w_3 + \dots x_nw_n + b$

$= \sum_{i=1}^{n} x_i w_i + b =$

$x_i =$

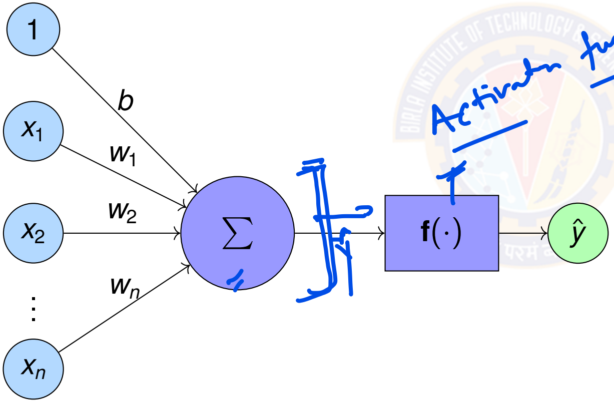
$w_i =$

$b =$

Artificial Neuron – Mathematical ComputaTION

- Sum of weights w_i times corresponding features x_i
- Apply activation function $f()$ on the sum.

Activation fun = End of 2nd class
3rd session



$$\boxed{z} = \sum_{i=1}^{n} w_i x_i + b \qquad (1)$$

$$\hat{y} = \mathbf{f}\left(\sum_{i=1}^{n} w_i x_i + b\right) \qquad (2)$$

Comparison: Biological vs Artificial Neuro

Biological Neuron

- Complex biochemical processes
- Analog signal processing
- Adaptive synaptic strengths
- Parallel processing
- Learning through experience
- Fault tolerant

Artificial Neuron

- Simple mathematical model
- Digital computation
- Adjustable weights
- Parallel computation possible
- Learning through algorithms
- Deterministic behavior

Key Insight

Artificial neurons are simplified mathematical abstractions that capture the essential computational properties of biological neurons.

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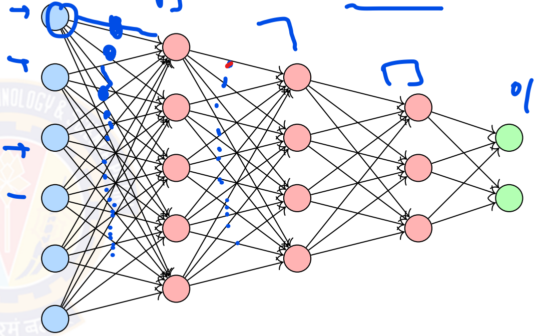
What are Artificial Neural Networks?

- Neural Network is a computational model inspired by human brain.
- Interconnected neurons process information.
- Interconnection can be forward, recursive or self loop.
- Learn patterns from data through training.
- Capable of approximating complex functions



Properties of Artificial Neural Networks

- Many neurons.
- Many weighted interconnections among neurons.
- Highly parallel, distributed process.
- Tuning parameters or weights automatically.

When to consider Artificial Neural Network

- Input is high-dimensional discrete or real-valued (e.g. raw sensor input).
- Possibly noisy data. Data has lots of errors.
- Output is discrete or real valued or a vector of values
- Form of target function is unknown.
- Human readability, in other words, explainability, of result is unimportant.
- Examples:
  - Speech recognition
  - Image classification
  - Financial prediction

4yrs sem

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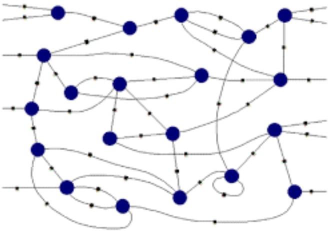
Connectionist Machines

- Network of processing elements, called artificial neurons.
- The neurons are interconnected to form a network.
- Interconnection can be forward, recursive or self loop.
- All world knowledge is stored in the connections between these elements.
- Neural networks are Connectionist machines.



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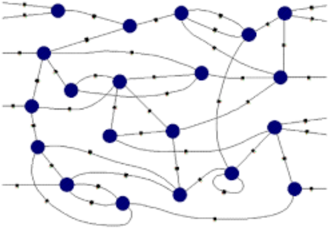
Connectionism Principles

Core Ideas:

- Intelligence emerges from simple processing neurons
- Knowledge is stored in connection weights
- Learning modifies connection strengths
- Parallel distributed processing



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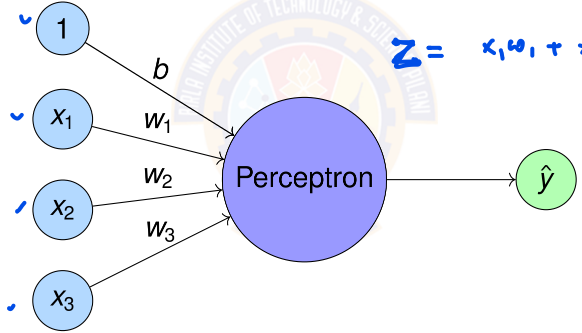
The Perceptron

Perceptron = Simplest form of artificial neuron.

Perceptron was introduced by Rosenblatt in $\boxed{1958.}$ old

$Z = x_1 w_1 + x_2 w_2 + x_3 w_3 + b$



$Z \begin{cases} > 0 \text{ — C.L} \\ < 0 \rightarrow \text{C.L} \end{cases}$

The Perceptron

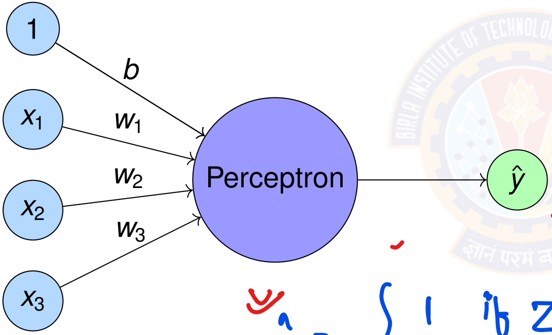
Binary classification

$Z > 0$

$Z < 0$

$Z = \sum_{i=1}^{n} w_i x_i + b$

Perceptron = Simplest form of artificial neuron.



$$\hat{y} = \begin{cases} \boxed{1} & \text{if } \sum_{i=1}^{n} w_i x_i + b \geq 0, \ Z \geq 0 \\ \boxed{-1} & \text{otherwise} \quad Z < 0 \end{cases}$$

↳ condition

$$\hat{y} = \begin{cases} 1 & \text{if } Z \geq 0 \\ 0 & Z < 0 \end{cases}$$

$w_1 = 1 \quad w_2 = 1$

$w_3 = 1 \quad b = -1.5$

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$\begin{array}{c} x_1 \\ x_2 \\ x_3 \end{array} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ input

$z = w_1x_1 + w_2x_2 + w_3x_3 + b$

$y =$

$z = 1\times1 + 1\times0 + 1\times1 - 1.5$

$z = 2 - 1.5 = \boxed{0.5} > 0$

$\boxed{\hat{y} = 1}$

$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ $\boxed{\hat{y} = -1}$

$1\times0 + 1\times1 + 0\times1 - 1.5$

$z_1 = -0.5 < 0$

y

Representing Logic Gates using Perceptron

- Power of Perceptron
  - Any linear decision can be represented by a Boolean equation.
  - The quest is to find out how to represent the above using Perceptron.
  - For this we test whether each of the logic gates can be represented by a Perceptron.
- For the perceptron algorithm:
  - The classical perceptron (Rosenblatt's original) typically uses {-1, +1} as outputs.
  - However, modern implementations and logic gates often use {0, 1}.
  - Both are valid! The choice affects the bias/threshold calculation.

Note: The numerical examples that follow uses binary values {0,1}. Classical perceptron uses {-1,+1}, but both formulations are equivalent.

AND Logic Gate

Question:

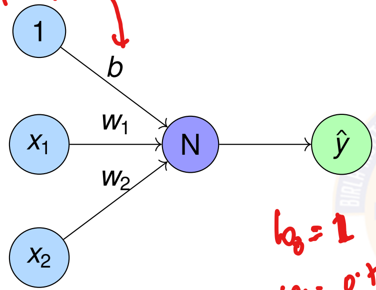
- How to represent AND gate using a Perceptron?
- What are the parameters for the AND Perceptron?
- Data is given below.

A	B	AB
0	0	0
0	1	0
1	0	0
1	1	1

Perceptron for AND gate

different weight

$b_0 = 1$



$b_0 = 1$

$w_1 = 0.75$

$w_2 = 0.75$

Goal: Find parameters w_0, w_1, w_2

x_1	x_2	t	h
0	0	0	$w_0 + 0 + 0 < 0$
0	1	0	$w_0 + 0 + w_2 < 0$
1	0	0	$w_0 + w_1 + 0 < 0$
1	1	1	$w_0 + w_1 + w_2 \geq 0$

One solution is

How

$$\boxed{\begin{aligned} w_0 &= -1 \\ w_1 &= 0.75 \\ w_2 &= 0.75 \end{aligned}}$$

← b_0

Present bias term

Perceptron equation is

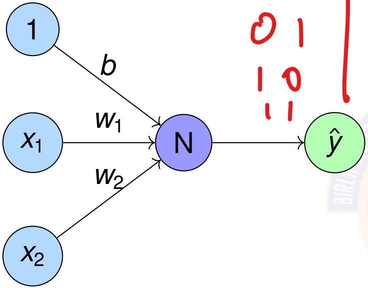
$\hat{y} <^{1}_{0}$ $\quad z > 0$ $\quad z < 0$

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2$$

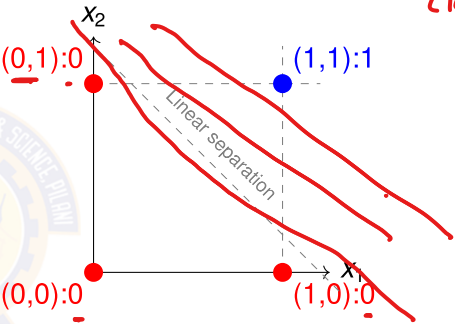
$\hat{y} = w_0 \times b + w_1 x_1 + w_2 x_2$

$\hat{y} = b + 0.75 x_1 + 0.75 x_2$

Perceptron for AND gate



0	0	0
0	1	0
1	0	0
1	1	1



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1

Perceptron equation is

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2$$

One solution is

$$w_0 = -1$$
$$w_1 = 0.75$$
$$w_2 = 0.75$$

OR Logic Gate

H.Q

Question:

- How to represent OR gate using a Perceptron?
- What are the parameters for the OR Perceptron?
- Data is given below.

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

$\boxed{\begin{array}{ll} z > 0 & \hat{y} = 1 \\ z < 0 & \hat{y} = 0 \end{array}}$

$w_1 = -1$

$w_2 = 2$

$b = 1.5$

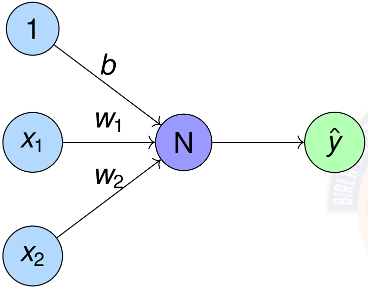
$0 + 1.5 < 0$

$1.5 < 0$

$-1 \times x_1 + 2 x_2 + 1.5 < 0$

fail $1.5 < 0$

Perceptron for OR gate



Goal: Find parameters w_0, w_1, w_2

x_1	x_2	t	h
0	0	0	$w_0 + 0 + 0 < 0$
0	1	1	$w_0 + 0 + w_2 \geq 0$
1	0	1	$w_0 + w_1 + 0 \geq 0$
1	1	1	$w_0 + w_1 + w_2 \geq 0$

One solution is

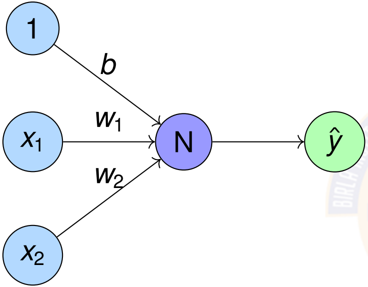
$$\boxed{\begin{aligned} w_0 &= -1 \\ w_1 &= 2 \\ w_2 &= 2 \end{aligned}}$$

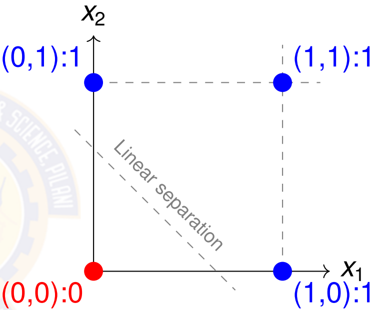
$b = x_0$

Perceptron equation is

$$\hat{y} = \underbrace{w_0 x_0}_{=} + w_1 x_1 + w_2 x_2$$

Perceptron for OR gate





Perceptron equation is

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2$$

One solution is
$$w_0 = -1$$
$$w_1 = 2$$
$$w_2 = 2$$

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y $\hat{y}$ Same?

y $\hat{y}$ Same

y $\hat{y}$ Same

misclassified

Perceptron Learning Algorithm

Goal: Find weights that correctly classify training data (input, output) pair (x, t).

Algorithm:

1. Initialize learning rate $\eta = 0.1$
2. Initialize weights randomly: $w_i = \text{random}$
3. For each training example (x, t):
   - Calculate output: $\hat{y} = sign(\sum_i w_i x_i + b)$
   - If $\hat{y} \neq t$, update weights:

$$w_i^{new} = w_i^{old} + \eta(t - \hat{y})x_i$$

$$b^{new} = b^{old} + \eta(t - \hat{y})$$

4. Repeat until convergence or max iterations

$W^{new} = W^{old} + \eta(t-\hat{y})x_i$

Actual value

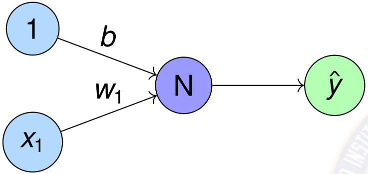
Predict

input

Convergence Theorem:

If linearly separable data, Perceptron learning algorithm will converge in finite steps.

Perceptron Learning for NOT gate

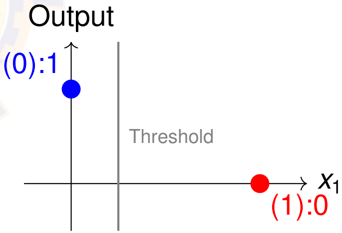


Example	x_1	t
Eg:1	0	1
Eg:2	1	0

Perceptron equation

$$h = w_0 x_0 + w_1 x_1$$

$$\hat{y} = \begin{cases} 1 & \text{if } h \geq 0 \\ 0 & \text{if } h < 0 \end{cases}$$



Perceptron Learning for NOT Gate I

1. Initialization
   - Assume $w_0 = w_1 = 0$.
   - Let the learning rate$= \eta = 1$.
2. Epoch 1
   - Example 1: $(x_1 = 0, t = 1)$

$$h = w_1 x_1 + w_0 = 0 \cdot 0 + 0 = 0$$

$$\hat{y} = \begin{cases} 1 & \text{if } h \geq 0 \\ 0 & \text{if } h < 0 \end{cases} = 1$$

Is $\hat{y}$ equal to t? YES. No weight update needed.

Perceptron Learning for NOT Gate II

- Example 2: $(x_1 = 1, t = 0)$

$$h = w_1 x_1 + w_0 = 0 \cdot 1 + 0 = 0$$
$$\hat{y} = 1$$

Is $\hat{y}$ equal to t? ($\hat{y} = 1, t = 0$) NO. Update weights:

$$w_i^{(new)} = w_i^{(old)} + \eta(t - \hat{y})x_i$$
$$w_1^{(new)} = 0 + 1(0 - 1) \cdot 1 = -1$$
$$w_0^{(new)} = 0 + 1(0 - 1) = -1$$

Perceptron Learning for NOT Gate III

3. Epoch 2 We will use the updated weights $w_1 = -1$, $w_0 = -1$.
   - Example 1: $(x_1 = 0, t = 1)$

$$h = w_1 x_1 + w_0 = (-1) \cdot 0 + (-1) = -1$$
$$\hat{y} = 0 \quad (\text{since } h < 0)$$

Is $\hat{y}$ equal to t ($\hat{y} = 0, t = 1$)? NO. Update weights:

$$w_1^{(new)} = (-1) + 1(1 - 0) \cdot 0 = -1$$
$$w_0^{(new)} = (-1) + 1(1 - 0) = 0$$

Perceptron Learning for NOT Gate IV

- Example 2: $(x_1 = 1, t = 0)$

$$h = (-1) \cdot 1 + 0 = -1$$
$$\hat{y} = 0$$

Is $\hat{y}$ equal to t? YES. No update needed. Current weights are $w_1 = -1, w_0 = 0$

4. Epoch 3

- Example 1: $(x_1 = 0, t = 1)$

$$h = (-1) \cdot 0 + 0 = 0$$
$$\hat{y} = 1 \quad \text{(since } h \geq 0\text{)}$$

Is $\hat{y}$ equal to t? YES. No update needed.

Perceptron Learning for NOT Gate V

- Example 2: $(x_1 = 1, t = 0)$

$$h = (-1) \cdot 1 + 0 = -1$$
$$\hat{y} = 0$$

Is $\hat{y}$ equal to t? YES. No update needed.

5. Algorithm Converges!
Final weights: $w_1 = -1, w_0 = 0$

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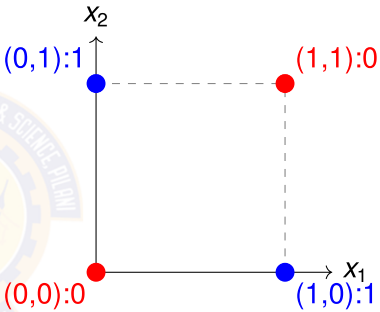
The XOR Problem

Truth Table for XOR:

x_1	x_2	x_1 XOR x_2
0	0	0
0	1	1
1	0	1
1	1	0



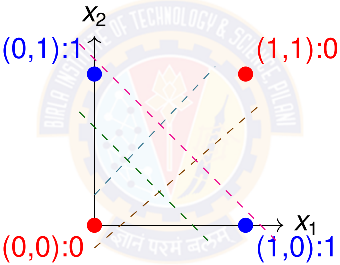
PROBLEM

XOR is not linearly separable! Single perceptron cannot solve XOR.

Why Single Perceptron Fails on XOR

Linear Decision Boundary Limitation:

Any line $w_1 x_1 + w_2 x_2 + b = 0$ cannot separate XOR classes.



No single line works!

Solution: Need multiple layers (Multilayer Perceptron)

session-3

Perceptron Demo Python Code

- https://colab.research.google.com/drive/1DUVcOoUIWhl8GQKc6AWR1wiOLaMeNkgD?usp=sharing

Student pl note:
The Python notebook is shared for anyone who has access to the link and the access is restricted to use BITS email id. So please do not access from non-BITS email id and send requests for access. Access for non-BITS email id will NOT be granted.

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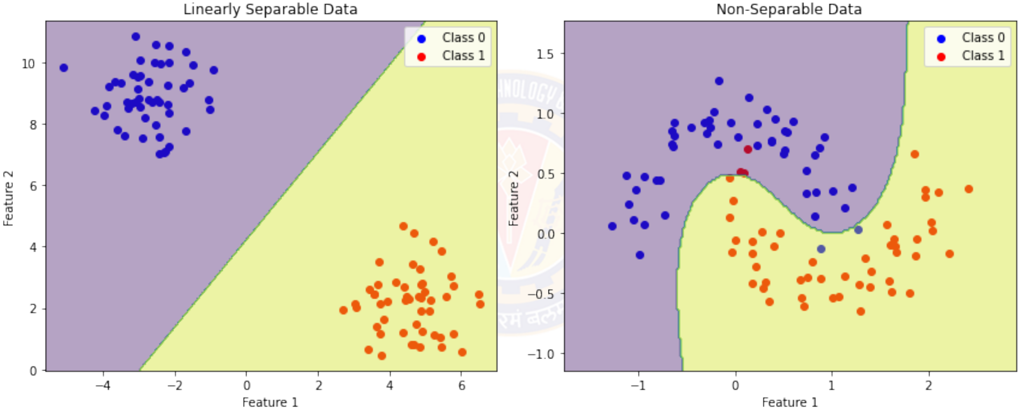
Linearly Separable Data

- Two sets of data points are linearly separable in a n-dimensional space if they can be separated by an $(n-1)$ dimensional hyperplane.
- in 2D-space, straight line can separate all the data examples belonging to class +1 from all the examples belonging to the class -1. Then it is 2D linearly separable data.
- An infinite number of straight lines can be drawn to separate the class +1 from the class -1.

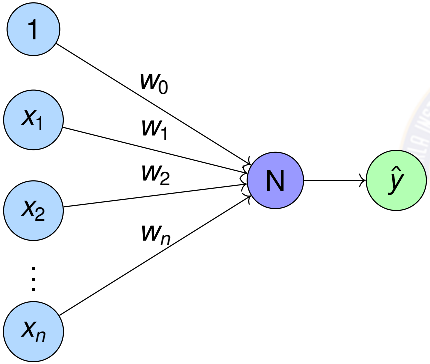
Linearly vs Non-Linearly Separable Data



Perceptron for Linearly Separable Data

- Data
  - Truth tables or set of examples
- Model
  - Perceptron
  - Inputs x are the features or columns
  - Target output t is the required label
- Objective Function
  - Deviation of desired output or target t and the computed output $\hat{y}$
- Learning Algorithm
  - Perceptron Learning algorithm
- Challenge
  - How to learn these $n+1$ parameters $w_0, w_1 . w_2 \ldots w_n$?
- Solution
  - Use Perceptron learning algorithm.

Perceptron for Linearly Separable Data



$$\tilde{x} = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \qquad w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$

$$\begin{aligned} h &= w_0 + w_1 x_1 + w_2 x_2 + \ldots + w_n x_n \\ &= w^\top \tilde{x} \end{aligned}$$

$$\hat{y} = \begin{cases} 1 & \text{if } h \geq 0 \\ 0 & \text{if } h < 0 \end{cases}$$

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Key Concepts

- Neural Network: Computational model inspired by biological neurons that processes information through interconnected units
- Perceptron:
  - Single-layer linear classifier: $\hat{y} = \text{sign}(w^\top x)$
  - Learning rule: $w_i^{(new)} = w_i^{(old)} + \eta(t - \hat{y})x_i$
  - Converges for linearly separable data
- Perceptron Limitations:
  - Cannot solve XOR and other non-linearly separable problems
  - Solution: Multi-layer networks with non-linear activation functions
- Foundation: These concepts form the basis for deep learning architectures

Practice Problems

1. Represent OR gate using Perceptron. Compute the parameters of the perceptron using perceptron learning algorithm.
2. Represent AND gate using Perceptron. Compute the parameters of the perceptron using perceptron learning algorithm.
3. Represent NOR gate using Perceptron. Compute the parameters of the perceptron using perceptron learning algorithm.
4. Represent NAND gate using Perceptron. Compute the parameters of the perceptron using perceptron learning algorithm.

- Zhang et al. ”Dive into Deep Learning” - Chapter 1 https://d2l.ai/
- http://mlsp.cs.cmu.edu/people/rsingh/docs/Chapter1_Introduction.pdf
- Goodfellow, Bengio, Courville - ”Deep Learning” Chapters 1-6
- Rosenblatt, F. (1958) - ”The Perceptron: A Probabilistic Model”
- Minsky & Papert (1969) - ”Perceptrons” (Historical perspective)

Thank You!